Neuron basics & communication: dendrites (input) → soma → axon (output); within-neuron signaling is an electrical spike (action potential) while between-neuron signaling is a chemical synapse; Brain vocabulary: gray matter = cell bodies, white matter = axons, the two hemispheres are linked by the corpus callosum, and cortex has frontal, parietal, temporal, and occipital lobes; Three brain methods & fMRI assumptions/limits: EEG is fast with low spatial detail, fMRI has good spatial detail but is indirect/slow BOLD, lesions/stimulation are causal but limited, and fMRI assumes blood-oxygen changes track local neural activity; Converging evidence: strong claims use multiple methods that point to the same conclusion; Hippocampus & memory (how we know): needed for forming new explicit memories (facts/events) as shown by patient cases like HM, imaging during memory tasks, and lesion evidence; Plasticity & LTP: plasticity means the brain changes with use, and LTP is repeated activity strengthening synapses (a cellular basis for learning); Other hippocampal roles & “binding”: spatial maps, context, and sequence/time support binding elements of experience in time and space; Habitual vs deliberative decisions: habitual is fast/automatic while deliberative is slower and plans/compares options; Reward prediction error (RPE) & dopamine: RPE = actual − expected reward, dopamine signals RPE, and basal ganglia support habit learning; Deliberation experiment (example): use tasks that force planning/tradeoffs and measure choices, reaction times, and neural signals during evaluation; Applications: habit change, addiction treatment, decision aids, and reinforcement-learning algorithms in AI; Everyday vs cosmic scales: we live at slow speeds/small energies while the universe spans extreme speeds/sizes; Constant-c puzzle & Einstein’s fix: light’s speed is always c and Einstein fixed this by letting space and time adjust (not c); Simultaneity: events that are same-time for one observer may not be for a moving observer; Time & space at high speed (when to use SR): time dilates and lengths contract with $\gamma = 1/\sqrt{1-(v/c)^2}$, use SR when v is a big fraction of c; $E = mc^2$ & moving fast: mass–energy equivalence and total energy at speed is γmc^2; Newton’s gravity limits: works for weak gravity/low speed but not near massive bodies or for light; GR in plain words: mass/energy curve spacetime and motion follows those curves (geodesics); Equivalence principle: free fall ≈ weightlessness locally and gravity is indistinguishable from acceleration; Evidence for GR: light bending, gravitational redshift, GPS corrections, Mercury’s perihelion shift, and gravitational waves; Black holes & time: Schwarzschild radius $r_s = 2GM/c^2$, near mass clocks run slower (gravitational time dilation), and r_s scales with mass; Expanding universe & Hubble data: farther galaxies recede faster (redshift ∝ distance) and the slope tells expansion rate/history; Double-slit outcomes & meaning: pellets give two bands, waves give interference fringes, electrons give fringes (wave-like) but measuring the path kills fringes; Probability in QM: outcomes are inherently probabilistic and the wavefunction gives probabilities; Uncertainty principle: you can’t know position and momentum precisely at once, with $\Delta x \cdot \Delta p \gtrsim \hbar/2$.

1 m = 100 cm = 1000 mm; 1 cm = 10 mm; 1 mm = 0.1 cm = 1×10^{-3} m; 1 µm (micrometer) = 1×10^{-6} m = 1×10^{-4} cm = 0.001 mm; 1 nm (nanometer) = 1×10^{-9} m; 1 cm² = $(1\times10^{-2}\text{ m})^2 = 1\times10^{-4}$ m²; 1 mm² = $(1\times10^{-3}\text{ m})^2 = 1\times10^{-6}$ m²; 1 µm² = $(1\times10^{-6}\text{ m})^2 = 1\times10^{-12}$ m²; 1 cm³ = $(1\times10^{-2}\text{ m})^3 = 1\times10^{-6}$ m³; 1 mm³ = $(1\times10^{-3}\text{ m})^3 = 1\times10^{-9}$ m³; 1 µm³ = $(1\times10^{-6}\text{ m})^3 = 1\times10^{-18}$ m³ = 1×10^{-12} cm³; Key used in 2C: 10^3 µm³ = 10^{-9} cm³; m/s → km/h: multiply by 3.6; m/s → mph: multiply by 2.237; km/h → m/s: divide by 3.6.

Lorentz factor: gamma = 1/sqrt(1 - (v/c)^2) — tells how strong relativistic effects are (time slows, lengths contract) at high speed; use when speed is a big fraction of light; v (object speed) comes from the problem, often as v/c (e.g., 0.99), and c is the speed of light (~3.0e8 m/s, but you usually just need the ratio v/c) ; Earth time: t_E = distance/speed — gives the trip time as measured on Earth; use when you know how far and how fast; distance (often in light-years) and speed (maybe as a fraction of c) are given in the prompt ; Shortcut in ly & c: t_E_years = distance_ly / (v/c) — skips unit conversions because 1 ly / c = 1 year; use when distance is in light-years and speed is given as a fraction of c ; Ship time: tau = t_E / gamma — gives how long the traveler experiences onboard; use after you have t_E and gamma; t_E from the previous step, gamma from the Lorentz formula ; Supplies: tubes = (tubes_per_year) * tau — converts ship time to how many items to bring; use when a per-year consumption rate is given; tubes_per_year comes from the problem, tau from above ; Mass→energy fraction (fusion): fraction = E / (m * c^2) — tells what fraction of mass became energy (efficiency); use for star/fusion questions; E (energy released) and m (mass used) are given; compare the fraction to benchmarks (e.g., diesel ~5e-10) by powers of ten ; Log–log plot: label axes in powers of 10 to show multiplicative trends across huge ranges; use for galaxy mass vs black-hole mass; both masses come from the dataset ; Schwarzschild radius: r_s = 2 * G * M / c^2 — event-horizon radius of a non-spinning black hole; use when mass M is given; G is Newton’s constant, c is speed of light ; Density (treat BH as a sphere): V = (4/3) * pi * r_s^3 ; rho = M / V — rough average density inside r_s; use after computing r_s; compare rho to air (~1 kg/m^3) for scale ; Heisenberg uncertainty: Delta_x * Delta_v >= h / (4 * pi * m) — minimum speed uncertainty from a given position uncertainty; use when Delta_x and particle mass m are given; solve for Delta_v and convert units (e.g., to mph) if asked.

For each group: $SE = SD/\sqrt{N}$. ; df = N - 1, get t* (95% two-sided). ; CI = Mean ± t* × SE. ; A = total highlighted surface area (cm²) you estimate from the scale bar; T = cortical thickness (cm); f = fraction of cortex that is neuron somata (e.g., 0.03 for 3%); v_soma = volume of one neuron soma (cm³); Cortical volume under the highlight V_cortex = A × T (cm³); Total soma volume; V_soma = f × V_cortex; Neuron count N = V_soma / v_soma = (f × A × T) / v_soma

Welch t (two groups): compare group averages; Lorentz factor: near light effects; ship proper time = Earth time/γ: onboard time experienced; Mass→energy fraction ($E/(mc^2)$): mass energy efficiency; t_E = distance/speed: compute travel time; τ = t_E/γ: relativistic onboard time; fraction = $E/(m\cdot c^2)$: mass energy efficiency.

df	t*
1	12.706
2	4.303
3	3.182
4	2.776
5	2.571
6	2.447
7	2.365
8	2.306
9	2.262
10	2.228
11	2.201
12	2.179

13	2.160
14	2.145
15	2.131
16	2.120
17	2.110
18	2.101
19	2.093
20	2.086
21	2.080
22	2.074
23	2.069
24	2.064
25	2.060
26	2.056
27	2.052
28	2.048
29	2.045
30	2.042
∞	1.960